Beyond Quadratic Lyapunov Functions: Estimating Region of Attraction by Tensor Lyapunov Equations

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***Abstract*—This note presents a novel framework for estimating the region of attraction (ROA) in nonlinear dynamical systems using tensor Lyapunov equations (TLEs), introduced as higher order extensions of the classical Lyapunov equation. By constructing Lyapunov functions from TLE solutions and establishing a criterion for subset verification, this method systematically enlarges ROA estimations. Numerical experiments demonstrate various ROA shapes, reduced conservatism, and scalability to high-dimensional systems, highlighting the effectiveness and generality of the proposed approach.**

***Index Terms*—Network analysis and control, optimization, region of attraction (ROA), stability of nonlinear systems.**

I. Introduction

Understanding the stability of dynamical systems critically depends on their operating states, which often manifest as asymptotically stable equilibria in realistic systems, such as ecological networks [1], power grids [13], [18], and social systems [6]. To characterize the stability of a desired equilibrium, the region of attraction (ROA) is proposed as a global indicator [9]. The ROA defines a set of initial conditions from which trajectories converge to the equilibrium.

System's initial condition matters. Even when there exist stabilizing control regimes on the underlying dynamical systems, prior knowledge regarding ROA size and shape remains essential. As indicated in [4], trajectories are more likely to diverge and reach abnormal states if initial values are too far from the expected working state, even with controls applied. This reveals that effective control is not sufficient on its own; understanding the ROA provides necessary guarantee that control strategies operate effectively. In other words, ROA complements control by clarifying where stabilization regimes are feasible in practice. Beyond its utmost importance on power grid design [18], robot control [8], and traffic regulation [17], ROA can still fit itself in newly developed fields. It serves as an essential index on how strong the resilience of a complex network is [14], describing a permissible state-perturbation range for system recovery. ROA can also function as a control planner in shifting hybrid systems in robotics [16].

Despite the criticality of ROA for system safety, stability, and resilience, the exact calculation of the ROA for general systems is usually impossible, especially when they are nonlinear or high-dimensional. Zubov's theorem [28] gives a necessary and sufficient description of ROA using partial differential equations. However, analytically solving this equation is only tractable in very simple cases, and obtaining numerical solutions is practically difficult [7]. For this reason, most researchers focus on finding inner estimations of ROA, which are unavoidably conservative compared to the actual ROA.

Recent approaches for ROA estimation are mainly divided into Lyapunov-based methods and non-Lyapunov methods. The former estimate ROAs by obtaining Lyapunov functions either theoretically or numerically, while the latter mostly applies data-driven learning techniques. The most widely used approach in Lyapunov-based methods is the sum-of-squares (SOS) optimization [3], [24]. Originally designed for polynomial systems, SOS formulation transforms the ROA estimation problem into a tractable semidefinite programming problem. More recently, new methods have been proposed to estimate ROAs for networked nonlinear dynamical systems [29], [30], yielding a spherical subset of ROA of synchronization dynamics by applying analytical techniques from complex networks. While they theoretically guarantee safety of the ROA estimated, their results can be highly conservative when scaled to large systems. Alternatively, data-driven non-Lyapunov estimations utilize reservoir computing [5], deep learning [25], and reverse trajectory methods [26]. Despite their flexibility, these approaches generally cannot provide an ROA estimation with full confidence, leaving a possibility that some predicted points fall outside the actual ROA.

As discussed above, the ROA estimation remains a major challenge across disciplines. This note focuses on a theoretical estimation of the ROA for general nonlinear dynamical systems. Our main contributions are summarized as follows.

1) We introduce the tensor Lyapunov equation (TLE) framework as an extension of the classical Lyapunov equation. Based on the unique solution theorem of TLE, we propose a novel approach to construct a sequence of Lyapunov functions for general nonlinear systems.
2) We propose a practical procedure to find a series of ROAs for general nonlinear dynamical systems. A criterion to check the subset condition in ROA estimation theorems is presented as a lemma, providing a theoretical basis to maximize the ROA estimation within this framework.
3) Numerical examples demonstrate that our ROA estimations can capture various shapes and significantly outperform previous theoretical works. Furthermore, using higher order polynomials beyond quadratic forms as Lyapunov functions better accommodates high-dimensional dynamical systems.

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The rest of this note is organized as follows. Some preliminaries on tensor algebra and dynamical systems will be introduced in Section II. Our main theorems on tensor Lyapunov equations and ROA estimation will be expanded in Section III and IV, respectively. Section V provides two numerical examples, validating the effectiveness of our theoretical results. Conclusion and outlooks lie in Section VI.

II. Mathematical Preliminaries

A. Foundations of Dynamical Systems

We begin by introducing the basic notation used throughout this note. For indices $i, j \in \{1, \ldots, n\}$, the Kronecker delta δ_{ij} equals 1 if $i = j$, and 0 otherwise. A square matrix Z is Hurwitz, if all its eigenvalues have strictly negative real parts. The notation $\|\cdot\|_p$ $(1 \leqslant p \leqslant +\infty)$ denotes the l_p-norm of a vector or a matrix A, defined as $\|A\|_p = (\sum_{i,j=1}^{n} |A_{ij}|^p)^{\frac{1}{p}}$ if $p < +\infty$, and $\|A\|_\infty = \max_{i,j} |A_{ij}|$. The matrix spectral norm is defined as $\|\cdot\|_{\mathscr{S}} = \sup_{\|x\|_2=1} \|Ax\|_2$. The Kronecker product is denoted by $\otimes$, and satisfies the following property.

Lemma 1: If square matrices A and B have eigenvalues $\lambda_1, \ldots, \lambda_k$ and $\mu_1, \ldots, \mu_l$, respectively, then the Kronecker product $A \otimes B$ has eigenvalues $\lambda_i \mu_j$, where $i \in \{1, \ldots, k\}, j \in \{1, \ldots, l\}$.

A function $h : \mathbb{R}^n \to \mathbb{R}$ is *positive definite* if $h(x) \geqslant 0 \forall x$, and $h(x) = 0$ only when $x = 0$. It is *radially unbounded*, if $|h(x)| \to +\infty$ as $\|x\| \to +\infty$. The β-sublevel set of h is

$$M_h(\beta) \triangleq \{x \mid h(x) \leqslant \beta\}$$

and the strict β-sublevel set is $M_h^\circ(\beta) = \{x \mid h(x) < \beta\}$. In this context, β is referred to as the level value. Clearly, $M_h(\beta_1) \subseteq M_h(\beta_2)$ if $\beta_1 \leqslant \beta_2$.

Consider the n-dimensional dynamical system with an equilibrium point s

$$\dot{x} = f(x) \tag{1}$$

where the dynamical function $f : \mathbb{R}^n \to \mathbb{R}^n$ is continuously differentiable. A set $\mathcal{O} \subseteq \mathbb{R}^n$ is a (positively) invariant set, if $x_0 \in \mathcal{O}$ indicates $x(t; x_0) \in \mathcal{O} \forall t \geqslant 0$, where $x(t; x_0)$ represents the trajectory starting at initial point x_0. Let $e = x - s$ be the error vector. Since $f(s) = 0$, the error system is

$$\begin{aligned} \dot{e} &= f(x) - f(s) = Jf(s)e + R(e) \\ &= Ze + R(e) \end{aligned} \tag{2}$$

where $Z = Jf(s)$ is the time-invariant Jacobian matrix of f evaluated at s, and $R(e)$ is the corresponding nonlinear residual. Neglecting $R(e)$ yields the linearized system

$$\dot{\xi} = Z\xi \tag{3}$$

whose solution is expressed as $\xi(t; x_0) = e^{tZ} x_0$. A fundamental property for linear systems is stated in the following lemma.

Lemma 2 (See [9]): The origin of linear system (3) is asymptotically stable, iff there exist positive constants c and λ, such that

$$\|e^{tZ}\|_{\mathscr{S}} \leqslant ce^{-\lambda t}. \tag{4}$$

Definition 1 (ROA around s): If the equilibrium point s is asymptotically stable, then the ROA of system (1) around s is defined as

$$R_A(s) = \{x_0 \mid \lim_{t \to +\infty} x(t; x_0) = s\}.$$

The ROAs of (1) and (2) are essentially identical up to a translation, i.e., $\{e_0 + s \mid \lim_{t \to +\infty} e(t; e_0) = 0\} = R_A(s)$. Without loss of generality, we set $s = 0$ in the subsequent discussion.

Lemma 3 (See [10] and [27]): Suppose that $\dot{x} = f(x)$ has an equilibrium at the origin. If there exists an open, connected subset S containing 0 and a continuously differentiable function $V : \mathbb{R}^n \to \mathbb{R}$, such that

$$V(x) > 0, \quad \dot{V}(x) < 0 \quad \forall x \neq 0 \text{ in } S$$

then every invariant set contained in S is a subset of the ROA.

Definition 2 (Residual function): The function $\phi^{(p)} : \mathbb{R}^n \to \mathbb{R}$ is defined as

$$\phi^{(p)}(x) = \begin{cases} \frac{\|f(x+s) - f(s) - Zx\|_p}{\|x\|_p}, & x \neq 0 \\ 0, & x = 0. \end{cases} \tag{5}$$

This is called the *p-norm residual function* of the dynamical function f. Given that f is continuously differentiable, and $\phi^{(p)}$ is continuous for any p.

Assumption 1 (A1): The equilibrium point 0 in the linearized system (3) is asymptotically stable, and there exist constants $c, \lambda > 0$ satisfying $\|e^{tZ}\|_{\mathscr{S}} \leqslant ce^{-\lambda t}$.

Assumption 2 (A2): The dynamical function f in system (2) is continuously differentiable.

Assumption 3 (A3): $M^\circ_{\phi^{(p)}}(\beta)$ is always an open region for any $\beta > 0$ and $p \geqslant 1$.

B. Notations on Tensor Operations

To represent higher order polynomials and extend Lyapunov equations, we introduce the following basic tensor operations.

An m-order n-dimensional (real) tensor $T^{(m,n)}$ is a multidimensional array with its entries $T^{(m,n)}_{i_1 \cdots i_m} \in \mathbb{R}$, where each index $i_j \in \{1, \ldots, n\}$ [20]. The order m denotes the number of indices ($m \geqslant 2$), and the dimension n defines the range of each index. The infinity norm is given by $\|T^{(m,n)}\|_\infty = \max_{1 \leqslant i_1, \ldots, i_m \leqslant n} |T^{(m,n)}_{i_1 \cdots i_m}|$. Unless otherwise specified, we write $T^{(m,n)}$ as $T^{(m)}$ hereafter. A tensor $T^{(m)}$ is symmetric if it is invariant under any index permutation. An even order tensor $T^{(2m)}$ is called a *unit tensor*, denoted $I^{(2m)}$, if $T^{(2m)}_{ii \cdots i} = 1 \, \forall i$ and all other entries are 0.

For $x_1, \ldots, x_m \in \mathbb{R}^n$, the *tensor-vector multiplication* is $T^{(m)} x_1 \cdots x_m = \sum_{i_1, \ldots, i_m = 1}^{n} T^{(m)}_{i_1 \cdots i_m} x_{1,i_1} \cdots x_{m,i_m}$, where x_{j,i_j} denotes the i_jth entry of vector x_j. If all vectors are identical ($x_i = x$), then the expression simplifies to

$$T^{(m)} x^m = \sum_{i_1, \ldots, i_m = 1}^{n} T^{(m)}_{i_1 \cdots i_m} x_{i_1} \cdots x_{i_m}.$$

A tensor $T^{(m)}$ is positive (semi)definite, if m is even, and $T^{(m)} x^m > 0 (\geqslant 0)$ for any nonzero vector $x \in \mathbb{R}^n$.

For $A \in \mathbb{R}^{n \times n}$, the *mode-j tensor-matrix multiplication* [20], [23] is defined as $(T^{(m)} \circ_j A)_{i_1 \cdots i_m} = \sum_{\alpha=1}^{n} T^{(m)}_{i_1 \cdots i_{j-1} \alpha i_{j+1} \cdots i_m} A_{\alpha i_j}$. Define the operator $\circ_{\mathcal{S}}$

$$T^{(m)} \circ_{\mathcal{S}} A = \sum_{j=1}^{m} T^{(m)} \circ_j A. \tag{6}$$

This operator possesses the following property, which is straightforwardly proved.

Lemma 4: If $T^{(m)}$ is a symmetric tensor, then for any (not necessarily symmetric) matrix A, $T^{(m)} \circ_{\mathcal{S}} A$ is always an m-order symmetric tensor.

Finally, we define a tensor-matrix operator $\bar{\circ}$ to represent the coordinate transformation of a tensor. Given $T^{(m)}$ and an $n \times n$ matrix A, the entries of $T^{(m)} \bar{\circ} A$ are calculated as

$$[T^{(m)} \bar{\circ} A]_{i_1 \cdots i_m} = \sum_{j_1, \ldots, j_m = 1}^{n} T^{(m)}_{j_1 \cdots j_m} A_{j_1 i_1} \cdots A_{j_m i_m}. \tag{7}$$

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III. Tensor Lyapunov Equation

To analyze the stability of the origin for a linear dynamical system $\dot{x} = Ax$, we typically choose a quadratic candidate Lyapunov function $V = x^T Px$. Assuming that the origin is asymptotically stable, or equivalently that A is a Hurwitz matrix, P is determined by the classical Lyapunov equation

$$A^T P + PA = -Q \tag{8}$$

where Q is a given positive definite matrix. Equation (8) possesses a unique symmetric positive definite solution $P = \int_0^{+\infty} e^{tA^T} Q e^{tA} dt$ [19], [22]. Furthermore, it can be expressed in the vectorized form $(I \otimes A^T + A^T \otimes I)\mathrm{vec}(P) = \mathrm{vec}(Q)$, where $\mathrm{vec}(\cdot)$ is the vectorization operator that stacks the columns of a matrix into a vector.

This framework is extended here to a higher order setting by introducing the TLE. Its definition and theoretical solution are given below.

Definition 3 (TLE in tensor form): Given a matrix $Z \in \mathbb{R}^{n\times n}$ and an $2m$-order positive definite tensor $Q^{(2m)}$, the tensor equation

$$P^{(2m)} \circ_{\mathcal{S}} Z = -Q^{(2m)} \tag{9}$$

is called the *TLE* in tensor form, where $P^{(2m)}$ is to be solved.

Similar to the classical Lyapunov (8), (9) can be transformed into a system of linear equations (SLE) through vectorization. The coefficient matrix of this SLE depends on the vectorization rule for tensors. In this note, we utilize a lexicographical rule where $\mathrm{vec}(T^{(m)}) \in \mathbb{R}^{n^m}$ is constructed as

$$\begin{aligned}\mathrm{vec}(T^{(m)}) = \Big(&T^{(m)}_{11\cdots1}, T^{(m)}_{21\cdots1}, \ldots, T^{(m)}_{n1\cdots1}\\ &T^{(m)}_{12\cdots1}, T^{(m)}_{22\cdots1}, \ldots, T^{(m)}_{n2\cdots1},\\ &\cdots\cdots,\\ &T^{(m)}_{1n\cdots n}, T^{(m)}_{2n\cdots n}, \ldots, T^{(m)}_{nn\cdots n}\Big)^T.\end{aligned} \tag{10}$$

This ordering guarantees a one-to-one mapping between tensor entries and vector components. Consequently, (9) corresponds to an n^{2m}-dimensional SLE with its coefficient matrix characterized as follows.

Lemma 5: Under vectorization rule (10), the coefficient matrix of TLE holds the form

$$C_{2m,Z} = \sum_{j=1}^{2m} I_{n^{2m-j}} \otimes (Z^T \otimes I_{n^{j-1}}). \tag{11}$$

The vectorized form of the TLE is defined using the notation from Lemma 5.

Definition 4 (TLE in vectorized form): Given a matrix $Z \in \mathbb{R}^{n\times n}$ and an $2m$-order tensor $Q^{(2m)}$, the SLE

$$C_{2m,Z}\mathrm{vec}(P^{(2m)}) = \mathrm{vec}(Q^{(2m)}) \tag{12}$$

is called the TLE in vectorized form.

Theorem 1 (Solution to TLE): If Z is Hurwitz, then the TLE (9) admits a unique symmetric positive definite solution, given by

$$P^{(2m)} = \int_0^{+\infty} Q^{(2m)} \bar{\circ} e^{tZ} dt. \tag{13}$$

Proof: First, we prove that the integral on the right-hand side of (13) is convergent, and that $P^{(2m)}$ is positive definite. We consider an arbitrary entry $P^{(2m)}_{i_1\cdots i_{2m}} = \int_0^{+\infty} \sum_{k_1,\ldots,k_{2m}=1}^{n} Q^{(2m)}_{k_1\cdots k_{2m}} e^{tZ}_{k_1 i_1} \cdots e^{tZ}_{k_{2m} i_{2m}} dt$. For a Hurwitz matrix Z, every e^{tZ}_{ij} can be upper bounded by an exponential function $c_{ij}e^{-\mu t}$ for some constants $c_{ij}, \mu > 0$ [11]. This ensures that the integral is convergent, and the right-hand side of (13) is entrywise finite. Based on the definition of $\bar{\circ}$, $P^{(2m)}$ is obviously a symmetric tensor and also of order $2m$. For any nonzero vector $x_0 \in \mathbb{R}^n$, we have

$$P^{(2m)} x_0^{2m} = \int_0^{+\infty} Q^{(2m)} \xi(t; x_0)^{2m} dt.$$

Since $Q^{(2m)}\xi(t;x_0)^{2m} > 0$ for any t, the integral $P^{(2m)}x_0^{2m}$ is finite and strictly positive.

Next, we prove that (13) is a solution to TLE. All we need is to verify that $(P^{(2m)} \circ_{\mathcal{S}} Z)_{i_1\cdots i_{2m}} = -Q^{(2m)}_{i_1\cdots i_{2m}}$ for an arbitrary group of indices $\{i_1, \ldots, i_{2m}\}$. According to definitions of tensor multiplications, the left-hand side expands as

$$\begin{aligned}&(P^{(2m)} \circ_{\mathcal{S}} Z)_{i_1\cdots i_{2m}}\\ &= \sum_{\alpha=1}^{2m} \left(P^{(2m)}_{\alpha i_2\cdots i_{2m}} Z_{\alpha i_1} + \cdots + P^{(2m)}_{i_1\cdots i_{2m-1}\alpha} Z_{\alpha i_{2m}}\right)\\ &= \int_0^{+\infty} \sum_{k_1,\ldots,k_{2m}=1}^{n} Q^{(2m)}_{k_1\cdots k_{2m}} \left(\sum_{\alpha=1}^{2m} e^{tZ}_{k_1\alpha} Z_{\alpha i_1}\right) \cdots e^{tZ}_{k_{2m} i_{2m}} dt\\ &\quad + \cdots\cdots\\ &\quad + \int_0^{+\infty} \sum_{k_1,\ldots,k_{2m}=1}^{n} Q^{(2m)}_{k_1\cdots k_{2m}} e^{tZ}_{k_1 i_1} \cdots \left(\sum_{\alpha=1}^{2m} e^{tZ}_{k_{2m}\alpha} Z_{\alpha i_{2m}}\right) dt.\end{aligned} \tag{14}$$

Note that for any $j \in \{1, \ldots, 2m\}$, $\sum_{\alpha=1}^{2m} e^{tZ}_{k_j\alpha} Z_{\alpha i_j}$ is exactly the (k_j, i_j) entry of matrix $e^{tZ}Z = de^{tZ}/dt$. Then, (14) is further transformed into

$$\begin{aligned}&\int_0^{+\infty} \sum_{k_1,\ldots,k_{2m}=1}^{n} Q^{(2m)}_{k_1\cdots k_{2m}} \frac{d}{dt}\left(e^{tZ}_{k_1 i_1} \cdots e^{tZ}_{k_{2m} i_{2m}}\right) dt\\ &= \sum_{k_1,\ldots,k_{2m}=1}^{n} Q^{(2m)}_{k_1\cdots k_{2m}} (0 - \delta_{k_1 i_1} \cdots \delta_{k_{2m} i_{2m}}) = -Q^{(2m)}_{i_1\cdots i_{2m}}.\end{aligned} \tag{15}$$

This confirms that (13) is indeed a solution to TLE.

Finally, we prove that this solution is unique. *Lemmas 1* and *5* imply that the coefficient matrix of vectorized TLE is nonsingular with all its eigenvalues strictly negative. Thus, the SLE in (12) has a unique vector solution. Also note that (9) and (12) are essentially equivalent and share the same solution. Therefore, (13) is one and the only solution of (9) and (12) both. □

Remark 1: The proposed TLE framework serves as a direct generalization of the classical Lyapunov stability theory. When the tensor order is set to $2m = 2$, the tensor equation (9), the vectorized system (12), and the integral solution (13) reduce exactly to the standard Lyapunov equation and its associated properties. This recovery demonstrates that the classical framework is a specific instance of the higher order extension presented in this note.

IV. Main Results

In this section, we rigorously establish our main theorems for ROA estimation and propose a practical framework for nonlinear dynamical systems.

Theorem 2: Suppose that assumptions (A1)–(A3) hold for error system (2). Let tensor $P^{(2m)}$ be the solution to TLE

$$P^{(2m)} \circ_{\mathcal{S}} Z = -I^{(2m)} \tag{16}$$

and define $V(e) = P^{(2m)}e^{2m}$. If there exists a $\beta > 0$, such that

$$M_V^{\circ}(\beta) \subseteq M_{\phi^{(2m)}}^{\circ}(\gamma_1) \tag{17}$$

where $\gamma_1 = 1/(2m\|P^{(2m)}\|_\infty)$, then $M_V^{\circ}(\beta)$ is a subset of ROA.

Proof: Differentiating V along the trajectories of (2) gives

$$\begin{aligned}\dot{V}(e) &= P^{(2m)}\dot{e}e^{2m-1} + \cdots + P^{(2m)}e^{2m-1}\dot{e}\\ &= (P^{(2m)} \circ_{\mathcal{S}} Z)e^{2m} + 2mP^{(2m)}R(e)e^{2m-1}.\end{aligned}$$

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By applying Hölder's inequality to bound the latter term, we obtain

$$\dot{V}(e) \leqslant -I^{(2m)}e^{2m} + 2m\|P^{(2m)}\|_\infty \|R(e)\|_{2m}\|e\|_{2m}^{2m-1}$$
$$= \|e\|_{2m}^{2m}(-1 + 2m\|P^{(2m)}\|_\infty \phi^{(2m)}(e)).$$

This implies that $\dot{V} < 0$ for all $e \in M^\circ_{\phi^{(2m)}}(\gamma_1)$. Hence, V is a valid Lyapunov function for system (2) within the open region $M^\circ_{\phi^{(2m)}}(\gamma_1)$ according to (A3). Therefore, $M^\circ_{\phi^{(2m)}}(\gamma_1) \subseteq \{e|\dot{V}(e) < 0\}$.

If there exists such a $\beta > 0$ satisfying the subset condition (17), then $M^\circ_V(\beta) \subset \{e|\dot{V}(e) < 0\}$. Since $M^\circ_{\phi^{(2m)}}(\gamma_1)$ is an open connected set and $M^\circ_V(\beta) \subseteq \{e|\dot{V}(e) < 0\}$, $M^\circ_V(\beta)$ constitutes an invariant set of system (2). Therefore, $M^\circ_V(\beta)$ is indeed a subset of ROA according to *Lemma 3*. □

Remark 2: The choice of $I^{(2m)}$ in (16) ensures both theoretical optimality and computational tractability. In classical linear system theory, setting $Q = I$ in (8) minimizes the solution's spectral norm, under a standard normalization condition $Q \succeq I$. Extending this to TLEs, $I^{(2m)}$ systematically minimizes the tensor infinity norm $\|P^{(2m)}\|_\infty$. Since Theorem 2 bounds the ROA via $\gamma_1 = 1/(2m\|P^{(2m)}\|_\infty)$, this choice mathematically maximizes the theoretical estimation bound. Practically, altering the right-hand tensor yields only marginal geometric changes to the invariant sets. Because optimizing higher order tensor entries is computationally prohibitive, $I^{(2m)}$ provides an optimal, tractable bound without requiring complex nonconvex optimizations.

In the proof of Theorem 2, the sublevel set of Lyapunov function V must lie within $M^\circ_{\phi^{(2m)}}(1/(2m\|P^{(2m)}\|_\infty))$, which depends directly on the norm of TLE solution tensor $P^{(2m)}$. However, solving the TLE in either tensor or vectorized form becomes computationally intractable as the system dimension or tensor order increases. To mitigate this issue, we establish a more easy-to-verify criterion that avoids direct computation of $P^{(2m)}$, through bounding $\|P^{(2m)}\|_\infty$ with information only about Z.

Theorem 3: Suppose that (A1)–(A3) hold for error system (2) with the origin as an equilibrium point. Let tensor $P^{(2m)}$ be the solution to TLE (16), and $V(e) = P^{(2m)}e^{2m}$. If there exists a $\beta > 0$, such that

$$M^\circ_V(\beta) \subseteq M^\circ_{\phi^{(2m)}}(\gamma_2) \tag{18}$$

where $\gamma_2 = \lambda/n^{m+1}c^{2m}$, then $M^\circ_V(\beta)$ is a subset of ROA.

Proof: We first observe that

$$\|P^{(2m)}\|_\infty = \|\int_0^{+\infty} I^{(2m)}\bar{\circ}e^{tZ}dt\|_\infty$$
$$\leqslant \int_0^{+\infty} \|I^{(2m)}\bar{\circ}e^{tZ}\|_\infty dt.$$

Using the definition of the $\bar{\circ}$ operator, we have

$$\|I^{(2m)}\bar{\circ}e^{tZ}\|_\infty = \max_{1\leqslant i_1,\ldots,i_{2m}\leqslant n} |\sum_{k=1}^{n} e^{tZ}_{ki_1}\cdots e^{tZ}_{ki_{2m}}|$$
$$\leqslant \sum_{k=1}^{n}\prod_{l=1}^{2m}\max_{1\leqslant i_l\leqslant n}|e^{tZ}_{ki_l}| = \sum_{k=1}^{n}\max_{1\leqslant i_1\leqslant n}|e^{tZ}_{ki_1}|^{2m}$$
$$\leqslant n\max_{1\leqslant k,i_1\leqslant n}|e^{tZ}_{ki_1}|^{2m} = n\|e^{tZ}\|_\infty^{2m}.$$

The matrix norm equivalences give $\|e^{tZ}\|_\infty \leqslant \|e^{tZ}\|_2 \leqslant \sqrt{n}\,\|e^{tZ}\|_{\mathscr{S}}$. From (A1) and Lemma 2, we have $\|e^{tZ}\|_{\mathscr{S}} \leqslant ce^{-\lambda t}$ for some $c, \lambda > 0$. Consequently

$$\|P^{(2m)}\|_\infty \leqslant n^{m+1}c^{2m}\int_0^{+\infty} e^{-2m\lambda t}dt = \frac{n^{m+1}c^{2m}}{2m\lambda}. \tag{19}$$

The relationship (19) shows that $\gamma_2 \leqslant \gamma_1$, hence

$$M^\circ_{\phi^{(2m)}}(\gamma_2) \subseteq M^\circ_{\phi^{(2m)}}(\gamma_1).$$

Therefore, any $\beta > 0$ satisfying the criterion (18) also satisfies (17), which implies that $M_V(\beta)$ is a subset of ROA. □

Remark 3: Theorem 3 avoids explicit computation of $P^{(2m)}$, while it may obtain a more conservative estimation of ROA. For a fixed system dimension n, the estimated ROA from Theorem 3 shrinks exponentially as m increases. Furthermore, the constants c and λ may only be estimated numerically in general, since analytical evaluation of $\|e^{tZ}\|_{\mathscr{S}}$ is rarely tractable for an arbitrary Z.

To provide ROA estimations, Theorems 2 and *3* assume the existence of β satisfying the subset conditions (17) and (18), respectively. A natural question arises: when does such a β exist, and how can we maximize it to obtain the optimal ROA estimation? The following lemma provides an answer.

Lemma 6: Suppose that continuous functions $g, h : \mathbb{R}^n \to \mathbb{R}$ are nonnegative, and g is positive definite and radially unbounded. Assume that all nonempty strict sublevel sets of g and h are open regions in $\mathbb{R}^n$. Given a constant $\beta_1 > 0$, define

$$\beta_2 \triangleq \inf_{x\in\{x|h(x)\geqslant\beta_1\}} g(x) > 0 \tag{20}$$

then β_2 is the maximum level value such that $M^\circ_g(\beta_2) \subseteq M^\circ_h(\beta_1)$.

Proof: First we prove that $\beta_2 > 0$. Due to the radial unboundedness of g, the infimum in (20) is not attained at infinity. If we assume $\beta_2 = 0$, then there exists a bounded sequence of points $\{x_n\} \subseteq \{x|h(x) \geqslant \beta_1\}$ such that $\lim_{n\to+\infty} g(x_n) = 0$. Then, a convergent subsequence $\{x_{n_k}\} \to x_0$ exists. Due to the continuity of g, we have $g(x_0) = \lim_{k\to+\infty} g(x_{n_k}) = 0$. Note that $x_0 \neq 0$, which contradicts with the positive definiteness of g.

Then, we prove that β_2 ensures $M^\circ_g(\beta_2) \subseteq M^\circ_h(\beta_1)$. Suppose that there exists a point $x \in \mathbb{R}^n$ satisfying $g(x) < \beta_2$ while $h(x) \geqslant \beta_1$, i.e., $x \in \{x|h(x) \geqslant \beta_1\}$. By the definition of the infimum, however, $\beta_2 \leqslant g(x)$, which directly induces a contradiction.

Finally, we prove that β_2 is the maximum level value preserving the subset condition. Because g is continuous, $M_g(\alpha_1)$ is a strict subset of $M_g(\alpha_2)$ for any $\alpha_2 > \alpha_1 \geqslant 0$. Otherwise, if $M_g(\alpha_1) = M_g(\alpha_2)$, then g is discontinuous on the boundary of $M_g(\alpha_1)$. For any constant $\epsilon > 0$, the property of infimum guarantees the existence of a point $x_1 \in \{x|h(x) \geqslant \beta_1\}$, such that $\beta_2 \leqslant g(x_1) < \beta_2 + \epsilon$. Consequently, $M^\circ_g(\beta_2 + \epsilon) \not\subseteq M^\circ_h(\beta_1)$. □

Under assumptions (A1)–(A3), the Lyapunov function $V = P^{(2m)}e^{2m}$ and the residual function $\phi^{(2m)}$ naturally satisfy the conditions of g and h in Lemma 6, respectively. Therefore, finding the optimal β requires computing the infimum of V over the set $\{x|\phi^{(2m)}(x) \geqslant \gamma_1(\text{or } \gamma_2)\}$. On the other hand, since $V(x)$ is radially unbounded, this infimum search can only be restricted to a sufficiently large bounded region D in practice. Therefore, the strictly positive solution to optimization problem (OP)

$$\beta^* = \min \quad V(x)$$
$$\text{s.t.} \quad x \in \{x|\phi^{(2m)}(x) \geqslant \gamma_1(\text{or } \gamma_2)\} \cap D \qquad \text{(OP)}$$

always exists according to *Lemma 6*, and represents the maximum level value satisfying the subset condition (17) or (18).

This theoretical foundation directly yields a concrete corollary for ROA estimation.

Corollary 1: Suppose that assumptions (A1)–(A3) hold for the error system (2). Let the tensor $P^{(2m)}$ be the solution to TLE (16), and set $V(e) = P^{(2m)}e^{2m}$. Then, the strict sublevel set $M^\circ_V(\beta^*)$ is the best sublevel set estimation of ROA with respect to Lyapunov function V, where β^* is the solution to OP.

Building upon *Corollary 1*, these theoretical results coalesce into a systematic procedure for estimating the ROA of any nonlinear system $\dot{x} = f(x)$ near an equilibrium s. The procedure is organized as follows.

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1) First, extract the error system $\dot{e} = Ze + R(e)$ and verify that the linearized matrix Z is Hurwitz. Ensure that the residual function $\phi^{(2m)}$ satisfies assumption (A3). Select candidate even tensor orders $2m$. Quadratic functions ($m = 1$) typically suffice for low-dimensional systems, whereas higher order tensors provide flexibility in higher dimensions.
2) Second, evaluate the threshold parameter γ_i. Depending on computational constraints, one can solve the TLE to obtain $P^{(2m)}$ and compute γ_1 exactly according to Theorem 2. Alternatively, one can numerically approximate the bounds c and λ to calculate the conservative threshold γ_2 based on Theorem 3.
3) Finally, solve the OP to determine the optimal level value β^*. This step finalizes the ROA estimation as $R_A(s) = \{e + s \mid V(e) \leqslant \beta^*\}$.

Remark 4: Unlike traditional spherical or ellipsoidal bounds, ROA estimates generated by this tensor-based framework can adapt to the diverse and flexible shapes inherent to multivariate polynomials. Estimations derived from different tensor orders are not necessarily nested, offering complementary approximations that can fit in specific application circumstances.

V. Numerical Examples

In this section, we estimate the ROAs for two concrete dynamical systems.

A. Example 1: 2-D Robotic Control System

In robot manipulator control, the nonlinear dynamical equation of a one-link robot [21] is expressed as

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + D\dot{q} + g(q) = u \tag{21}$$

where q is a scalar denoting the generalized position coordinate, u is a control input, and $M(q)$ is a positive function representing the inertia at any position. The term $C(q,\dot{q})\dot{q}$ accounts for centrifugal and Coriolis forces, where the function $C = \frac{\partial M}{\partial q}\dot{q}/2$. The term $D\dot{q}$ accounts for viscous damping, where D is a positive constant; and the term $g(q)$ represents gravity forces. The asymptotic stability of the equilibria for system (21) under various control inputs has been well studied via Lyapunov methods [9]. In this note, we further estimate the ROA of its equilibrium.

The control input is defined as

$$u = g(q) - K_p(q - q^*) - K_d\dot{q} \tag{22}$$

where K_p and K_d are positive constants representing the control gains for the generalized position and velocity, respectively, and q^* is the desired robot position. Under this control law, the phase point $q = q^*, \dot{q} = 0$ becomes an asymptotically stable equilibrium. To emphasize the nonlinear nature of this system, $M(q)$ is chosen as e^q in this example, and $C(q,\dot{q}) = \frac{1}{2}e^q\dot{q}$. Parameters are set to $D = 0.1$ and $K_p = K_d = 5$, and the desired position is $q^* = 1$. The resulting linearized matrix is

$$Z = \begin{pmatrix} 0 & 1 \\ -1.84 & -1.88 \end{pmatrix}.$$

Its eigenvalues are $\lambda_{1,2} = -0.94 \pm 0.98i$, confirming it is a Hurwitz matrix. Fig. 1 shows that $\|e^{tZ}\|_{\mathscr{S}}$ is upper bounded by an exponential function $2.5e^{-0.93t}$, thus parameters in (A1) are selected as $c = 2.5$ and $\lambda = 0.93$.

The candidate tensor orders are chosen as 2, 4, and 6. Corresponding tensors $P^{(2)}, P^{(4)}$, and $P^{(6)}$ are acquired by applying Theorem 1. We denote the Lyapunov function formed by $P^{(2m)}$ as $V_{2m}(x) = P^{(2m)}x^{2m}$, where $m = 1, 2, 3$. Both Theorems 2 and 3 are applied to each tensor. After solving the OP for all cases, the best level values for V_2, V_4, and V_6 are summarized in Table I.

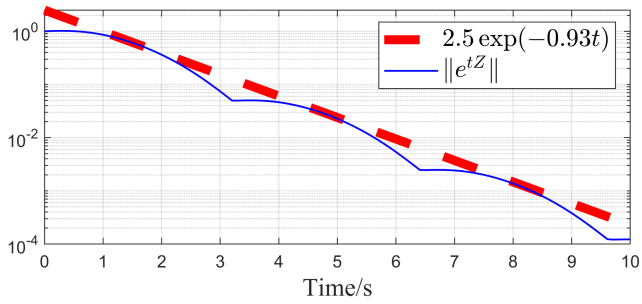


Fig. 1. Exponential upper bound for $\|e^{tZ}\|_{\mathscr{S}}$ in semilog coordinates.

TABLE I
Best Level Values for Lyapunov Functions of Different Tensor Orders Under Theorems 2 and 3

	Theorem 2	Theorem 3
V_2	$\beta_2^{(T2)} = 0.0377$	$\beta_2^{(T3)} = 2.79 \times 10^{-4}$
V_4	$\beta_4^{(T2)} = 3.23 \times 10^{-4}$	$\beta_4^{(T3)} = 2.79 \times 10^{-4}$
V_6	$\beta_6^{(T2)} = 3.18 \times 10^{-6}$	$\beta_6^{(T3)} = 4.91 \times 10^{-8}$

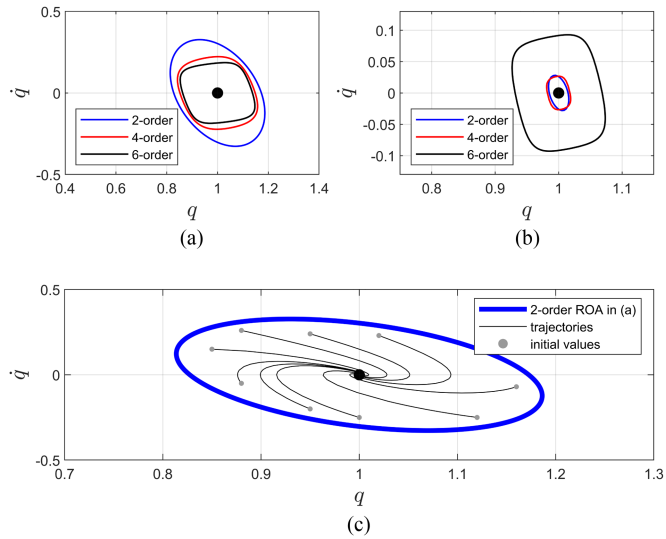


Fig. 2. Estimated ROA boundaries of Example 1, with tensor orders 2, 4, and 6, (a) using Theorem 2, and (b) using Theorem 3. (c) Trajectories from randomly chosen initial points inside the second-order ROA of (a), all converging to the equilibrium $(q^*, 0)$ (the big black point in all three panels).

From these results, six distinct ROAs are obtained, namely, $M^{\circ}_{V_{2m}}(\beta_{2m}^{(T2)})$ and $M^{\circ}_{V_{2m}}(\beta_{2m}^{(T3)})$, where $m = 1, 2, 3$. These are correspondingly called the two-, four-, and six-order ROAs for simplicity. The ROA estimations obtained by Theorems 2 and 3 are, respectively, shown and compared in Fig. 2(a) and (b). Coinciding with theoretical predictions, the ROAs in Fig. 2(b) are significantly smaller than those in (a). Furthermore, several initial values are randomly sampled within the two-order ROA of Fig. 2(a). Their trajectories are all attracted by the equilibrium point $(q^*, 0)$, as shown in Fig. 2(c).

B. Example 2: 8-D Networked System

The second example, adapted from [29], is an 8-D pendulum system defined on a four-node directed weighted network. The network

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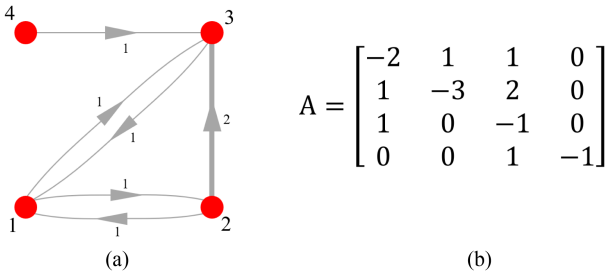


Fig. 3. (a) Structure of the directed weighted network in Example 2, where the number label of an edge indicates its weight. (b) Corresponding coupling matrix of the network.

structure is encoded by a coupling matrix A. Here, $A_{ij} = w_{ij}$ if there is a directed edge from node i to node j with weight w_{ij}, and $A_{ii} = -\sum_{j\neq i} A_{ij}$. The structure and its corresponding coupling matrix are shown in Fig. 3.

The system dynamics is given by

$$\dot{x}_i(t) = f(x_i(t)) + c\sum_{j=1}^{4} A_{ij}h(x_j(t)), \quad i = 1, 2, 3, 4 \tag{23}$$

where $x_i \in \mathbb{R}^2$ is the state vector of node i, and $c > 0$ is the coupling strength of the system. Each node follows identical dynamics $f : \mathbb{R}^2 \to \mathbb{R}^2$, indicating the evolution rule of an isolated node. Function $h : \mathbb{R}^2 \to \mathbb{R}^2$ specifies the inner coupling function, describing how the entries of x_i influence each other in their respective dynamics. Here, $c = 0.25$ and functions f and h maintain the same as in [29]

$$f(z) = (z_2, -\sin z_1 - 2z_2)^T, h(z) = (z_1 + 0.1\sin z_2, z_2)^T. \tag{24}$$

The origin of $\mathbb{R}^8$ is an equilibrium, and the largest real part of the eigenvalues of matrix Z is -1, confirming that Z is Hurwitz. We estimate the ROA of the origin and compare our results with those in [29]. Due to the high dimension of this example, we focus only on the results obtained by Theorem 2, as those by Theorem 3 would be overly conservative in this setting.

Candidate tensor orders 2, 4, 6, and 8 are considered, yielding tensors $P^{(2)}, P^{(4)}, P^{(6)}$, and $P^{(8)}$ solved by applying Theorem 1. These tensors subsequently generate Lyapunov functions V_2, V_4, V_6, and V_8 correspondingly. For each case, the OP is solved, resulting in the level values $\beta_2 = 1.33$, $\beta_4 = 2.06$, $\beta_6 = 3.10$, and $\beta_8 = 4.89$. Four distinct ROA estimations $M^{\circ}_{V_2}(\beta_2)$, $M^{\circ}_{V_4}(\beta_4)$, $M^{\circ}_{V_6}(\beta_6)$, and $M^{\circ}_{V_8}(\beta_8)$ are subsequently produced, which we refer to as two-, four-, six- and eight-order ROAs, respectively. A visualization of these ROA estimations is presented in Fig. 4, where the8-D ROAs are sectioned along multiple 2-D planes. Notably, the two-order ROA does not appear in Fig. 4(a) and (i), indicating that the corresponding planes do not intersect with this region. This observation highlights a significant enlargement of higher order ROAs compared with the quadratic case.

Initial values randomly sampled within each ROA can steer their corresponding trajectories to be exponentially attracted by the origin, as shown in Fig. 5. The numerical results validate the effectiveness of our ROA estimation procedure.

We also analyze the geometry of the estimated ROAs. First, $M^{\circ}_{V_2}(\beta_2) = \{x \mid x^T P^{(2)} x < \beta_2\}$ is an 8-D open ellipsoid. Its semi-axis lengths are $c_1 = 0.82, c_2 = 1.12, c_3 = 1.44, c_4 = 1.64, c_5 = 2.07, c_6 = 2.33, c_7 = 2.58$, and $c_8 = 2.73$. In contrast, the ROA estimation reported in [29] is a closed ball of radius 0.0906, significantly smaller than the smallest semiaxis length 0.82, meaning that it is strictly contained within our ellipsoid.

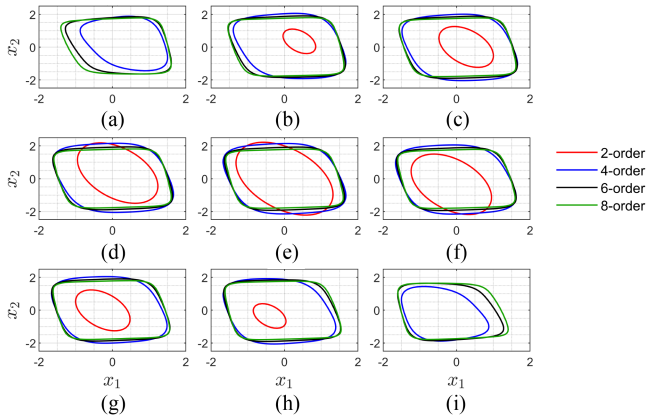


Fig. 4. 2-D ROA profiles on $x_1 - x_2$ coordinates, compared across different tensor orders. Rows correspond to fixed values of $x_5 = -1, 0, 1$ (top to bottom), and columns to fixed values of $x_6 = -1, 0, 1$ (left to right). All other coordinates are set to zero ($x_3 = x_4 = x_7 = x_8 = 0$).

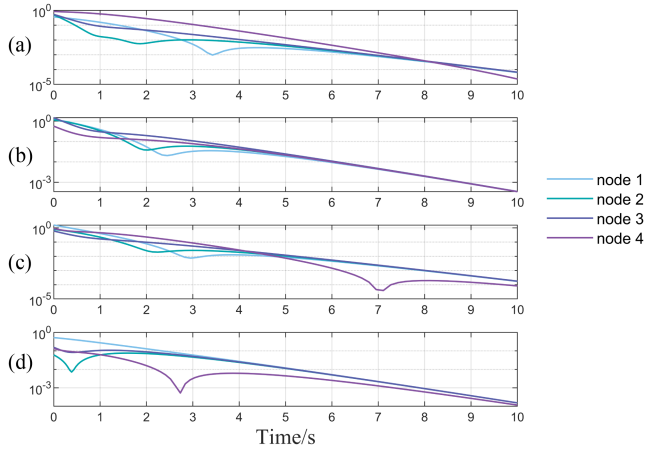


Fig. 5. Norms of the error vectors for the four nodes, plotted in semilog coordinates. Initial values are randomly extracted from ROAs obtained in the tensor order (a) 2, (b) 4, (c) 6, and (d) 8 situations, respectively.

For higher order Lyapunov functions, their ROA boundaries are much more complicated and cannot be analytically described. Nevertheless, they share a fundamental property that the nonempty strict sublevel sets of V_{2m} are always open regions and centrosymmetric with respect to the origin. To characterize these boundaries, therefore, we sample 10^6 uniformly distributed unit vectors, representing directions in $\mathbb{R}^8$. For each unit vector p, we compute the maximal distance it can be extended before intersecting the boundary of the set. This distance, referred to as the *set extension in direction* p, encodes information about the boundary points along that direction. Collecting the set extensions over all sampled directions yields a comprehensive approximation of the boundary geometry, enabling straightforward comparisons between different ROAs. In particular, if the number of sampled directions is sufficiently large, then the condition that the set extension in every direction of A is no greater than that of B provides a reliable *approximate criterion* for validating the inclusion $A \subseteq B$.

To obtain the (hyper)volume of each ROA, a Monte Carlo simulation is carried out. The estimated volume of $M^{\circ}_{V_2}(\beta_2)$ is 299.37, closely matching its analytical value of 300.73 with a relative error below 0.5%. The estimated volumes of $M^{\circ}_{V_4}(\beta_4)$, $M^{\circ}_{V_6}(\beta_6)$, and $M^{\circ}_{V_8}(\beta_8)$

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are 4610.88, 6106.43, and 6208.31, respectively. A further inclusion relationship detection reveals that $M^{\circ}_{V_2}(\beta_2)$ can be considered fully contained in $M^{\circ}_{V_4}(\beta_4)$. It is almost entirely contained in $M^{\circ}_{V_6}(\beta_6)$ and $M^{\circ}_{V_8}(\beta_8)$ in 99.6% and 98.9% of the sampled directions, respectively. Although Fig. 4(d)–(f) suggests that the two-order ROA is not in fact strictly contained in the four-order ROA, an approximate inclusion relationship is still valid. These discussions demonstrate that higher tensor orders yield consistently larger ROA estimates in high-dimensional systems.

VI. Conclusion and Outlooks

In this note, we have developed a novel theoretical framework to estimate the ROA of nonlinear dynamical systems. The TLE has been proposed as the core innovation, extending the classical Lyapunov equation. By exploiting TLE solutions, we have systematically constructed higher order Lyapunov functions, which have provided critical flexibility to address complex dynamics. Subsequently, optimal ROA estimation theorems have been rigorously established. Theorem 2 may be more time-consuming in complicated situations, but it acquires a far better ROA estimation compared to Theorem 3. Through validations on robotic and high-dimensional networked systems, our methodology has been effectively demonstrated. Our results have shown that this tensor-based approach has successfully captured diverse, nonellipsoidal boundaries and has significantly enlarged estimation scopes. Overall, this work has robustly integrated Lyapunov analysis with tensor formulations, which has advanced both the theoretical and practical frontiers of ROA estimation.

In future works, we plan to construct nonhomogeneous Lyapunov functions that mix multiple even-degree tensor terms to enhance the underlying bounding mechanism. In conjunction with this, generalizing the residual function beyond standard norms could tighten analytical bounds and yield nonspherical sets that more accurately map the underlying vector fields. To be applied on large-scale networked systems, TLEs can be decomposed into lower dimensional subsystems by exploiting network sparsity, dramatically improving computational efficiency. Ultimately, leveraging these estimations to design control regimes that actively enlarge a system's ROA remains a vital direction for exploration.

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